DAY8

Problem Statement

Automated Teller Machines (ATMs) are equipped with CCTV cameras to continuously record customer activities and ensure security. In the event of incidents such as robbery, ATM tampering, suspicious loitering, crowd formation, or unusual behavior, investigators are required to manually review large amounts of surveillance footage to identify relevant evidence. This manual investigation process is time-consuming, labor-intensive, and may lead to important events being overlooked.

To address this challenge, this project proposes an AI-Based ATM CCTV Surveillance Analysis and Incident Detection System that assists investigators in analyzing recorded ATM surveillance videos. The system automatically detects ATM machines and individuals, tracks visitor movements, identifies suspicious activities such as loitering, crowding, repeated ATM interactions, possible ATM tampering, and aggressive behavior, and extracts important incident clips for further investigation. In addition, the system generates structured investigation reports containing event timestamps, visitor statistics, and evidence clip information, thereby significantly reducing CCTV review time and improving investigation efficiency.

Dataset Collection and Annotation

A suitable dataset was prepared to support the development of the ATM CCTV Surveillance Analysis and Incident Detection System. The dataset consists of ATM surveillance videos containing both normal customer activities and suspicious events. To identify the occurrence of abnormal activities within the videos, temporal annotations were maintained using a label file.

The annotation file follows the structure:

Video_Name Total_Frames FPS Start_Frame End_Frame

Example:
10.mp4 1796 30 349 1796

In this example:
• 10.mp4 represents the video name.
• 1796 represents the total number of frames in the video.

• 30 represents the frame rate (Frames Per Second).
• 349 represents the frame at which suspicious activity begins.
• 1796 represents the frame at which the suspicious activity ends.

These annotations provide temporal information about abnormal events present in the surveillance footage. Using the start and end frame ranges, the suspicious segments of the video can be identified without manually reviewing the entire recording.

The annotation information was utilized during dataset preparation to separate normal and anomalous video segments. This enabled efficient analysis of surveillance footage and helped in training and evaluating the incident detection pipeline. The annotated frame ranges also support the extraction of evidence clips and the generation of investigation reports by identifying the exact portions of the video where suspicious activities occur.

Annotated Dataset Link:
https://drive.google.com/drive/folders/1xxLFNF3JAAGTDfVkO9vyqzjdUYMGeQxP?usp=sharing